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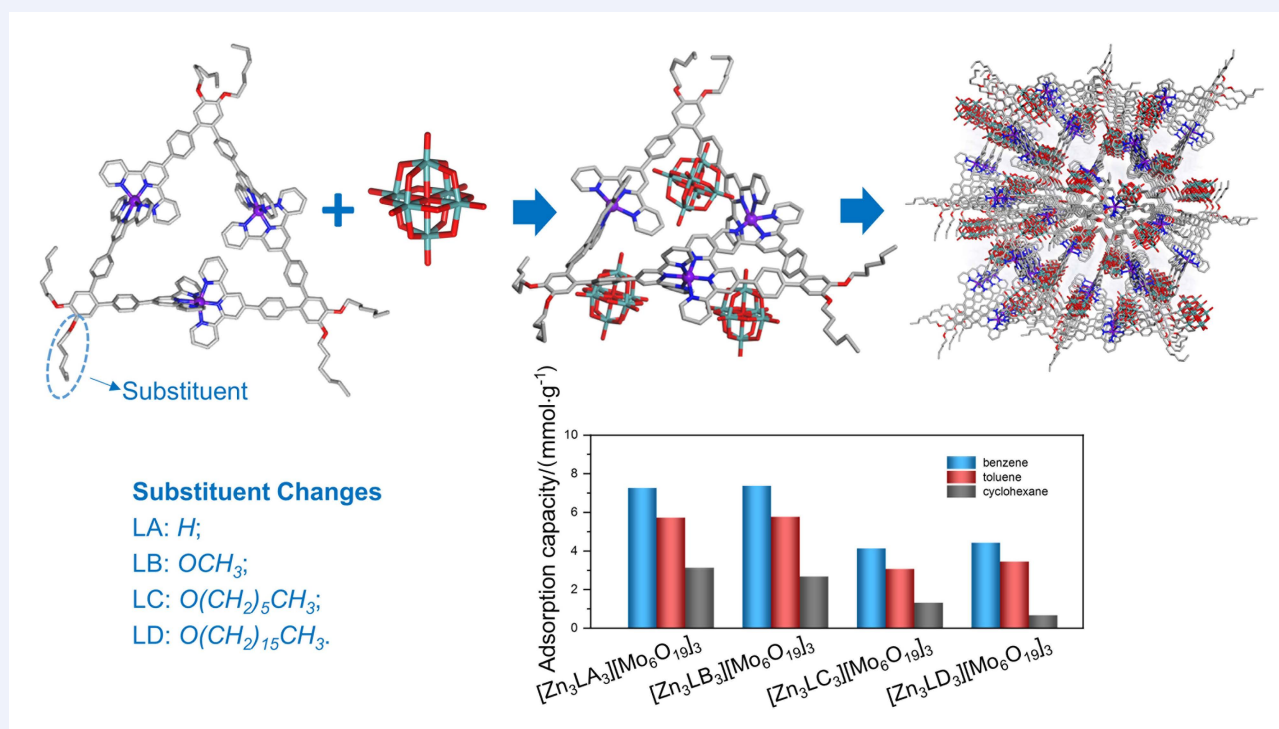
Porous Assembly of Metallo-Supramolecule and Polyoxometalate via Ionic Complexation with Vapor Sorption Properties

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Comprehensive Summary

Porous materials have been extensively developed and displayed special applications in diverse fields, however, it still remains a great challenge in developing new synthetic methods with features of energy-saving and easy-operating. In this study, we reported the construction of a series of novel ionic complexes based on discrete supramolecular triangle structure and polyoxometalate [Mo₆O₁₉]²⁻, using a quite efficient strategy. The obtained composites exhibit precise 3D network structures with ideal stability containing micropores, which were characterized by the ¹H NMR, ESI-MS, TEM, SEM, and single-crystal X-ray diffraction. These composites showed differential vapor adsorption behaviors for benzene, toluene, and cyclohexane. This work provides a novel approach for fabricating porous materials through electrostatic interactions and has great potential for gas separation.



Keywords

Self-assembly | Metallo-supramolecule | Polyoxometalates | Electrostatic interactions | Vapor sorption

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Background and Originality Content

Porous materials have attracted immense attention and found their potential in ion-exchange, reaction/catalysis, and chemical storage, adsorption and separation, due to their considerable surface area and unique pore structures.^[1–6] Representative porous materials, such as inorganic zeolites,^[7–8] metal-organic frameworks (MOFs),^[9–10] covalent organic frameworks (COFs),^[11–12] and porous aromatic frameworks (PAFs)^[13] have been synthesized mainly by hydrothermal and solvothermal methods,^[14–16] in which high temperature, high pressure, and long reaction time are usually required for the structural refinement in preparation of the final frameworks. Therefore, it is necessary to develop more energy-conserving and easy-operating methods to construct porous materials for industrial applications. In contrast to the coordination and dynamic covalent interactions typically used in preparing MOFs/COFs, electrostatic interactions only require mild reaction conditions and a short time to form a solid with a crystallographic lattice by stacking in an alternating fashion.^[17–18] However, the porous materials based on electrostatic interactions have rarely been reported, due to the uncontrollability of long-range Coulombic forces.^[19–20] Rapid and precise construction of porous materials through adjusting electrostatic interactions remains a great challenge.

Among the large variety of ionic components, supramolecular coordination complexes (SCCs) and polyoxometalates (POMs) have been receiving extensive attention.^[21–27] As a class of typical cationic skeleton, SCCs can provide rigid skeleton with precise structure and cation amount.^[28–30] Up to the present, a wide variety of 2D (such as triangles, rhomboids, rectangles, and hexagons) and 3D (cubes, tetrahedrons, and other cages) SCCs have been reported.^[31–34] POMs as a category of negative metal oxide cluster exhibit the advantages of well-defined structures and tunable charges.^[35–36] Due to the diversity and structural stability of SCCs and POMs, an ionic complexation assemblies' library can be established based on their electrostatic interactions.

Herein, we reported a strategy to fabricate porous materials by electrostatic interactions between SCCs and POMs. The metallo-triangles as cation building blocks have shown stability to accommodate anions at three vertices for the steric hindrance of pseudo-octahedral $\langle \text{tpy-Zn}^{2+}\text{-tpy} \rangle$ coordination moieties^[37] in the middle of each edge. To coordinate the shapes, sizes, and charge valences, the divalent octahedron POMs $[\text{Mo}_6\text{O}_{19}]^{2-}$ with a diameter of 0.8 nm was selected.^[38] The compositions and morphologies of these composites were investigated by NMR, ESI-MS, TEM, and SEM. In order to further clarify the material structure, the X-ray crystal structure of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ was obtained, which showed the expected 3D precise structure. With the existence of micropores, the ionic complexes showed adjustable vapor adsorption properties for benzene and cyclohexane by changing the length of alkyl chains of the metallo-triangles.

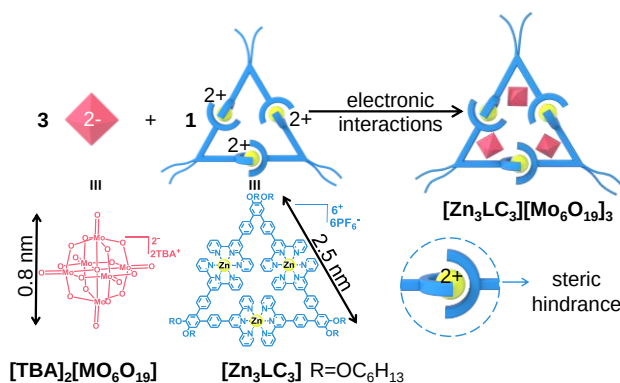
Results and Discussion

Synthesis of the composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$

Ligand **LC** and metallo-triangle $[\text{Zn}_3\text{LC}_3]$ with six positive charges (Scheme 1) were synthesized according to the previous literature.^[39] Charge balanced composite was obtained by adding an acetonitrile solution of $[\text{TBA}]_2[\text{Mo}_6\text{O}_{19}]$ with two negative charges dropwise into another acetonitrile solution of $[\text{Zn}_3\text{LC}_3]$ in a stoichiometric ratio (3 : 1). In this process, a yellow precipitate was immediately formed, followed by filtering and washing with acetonitrile to give the composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ (yellow power, yield 90%) without further purification. The obtained composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ is highly stable (Figure S30) and insoluble in common solvents, such as chloroform, acetonitrile,

acetone, tetrahydrofuran, and toluene, whereas it is only soluble in *N*-methyl-2-pyrrolidone (NMP) and slightly soluble in DMSO and DMF. In order to verify the charge neutralization of the final product, the $[\text{Zn}_3\text{LC}_3]$ and $[\text{TBA}]_2[\text{Mo}_6\text{O}_{19}]$ were mixed in a stoichiometric ratio of 1 : 1. After centrifugation, the ^1H NMR spectrum of the supernate showed that the ratio of residual $[\text{Zn}_3\text{LC}_3]^{6+}$ and $[\text{TBA}]^+$ is 1 : 3, suggesting the ratio of $[\text{Zn}_3\text{LC}_3]^{6+}$ and $[\text{Mo}_6\text{O}_{19}]^{2-}$ is 1 : 3 in the obtained composite (Figure S25).

Scheme 1 The synthesis of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$



The ^1H NMR spectra of ligand **LC**, supramolecule $[\text{Zn}_3\text{LC}_3]$, and composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ were shown in Figure 1. The ^1H NMR spectrum of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ (Figure 1c) was significantly broadened, suggesting the successful synthesis of the large composite. Compared with $[\text{Zn}_3\text{LC}_3]$, the Ph-a protons of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ showed a downfield shift from δ 8.44 to δ 8.47. While the tpy-3,3' protons in the electronic shielding area showed obvious upfield shift with 0.03, indicating the in-

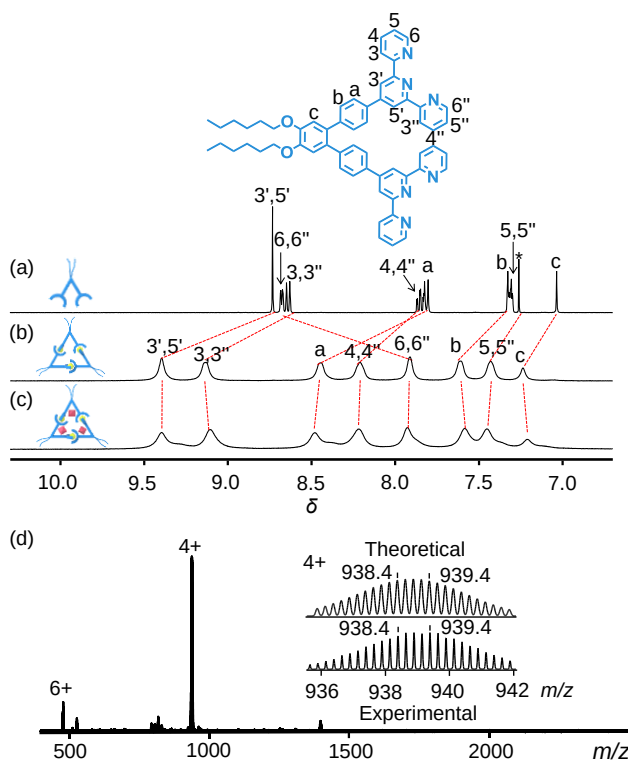


Figure 1 ^1H NMR spectra (500 MHz, 300 K) of (a) ligand **LC** in CDCl_3 , (b) $[\text{Zn}_3\text{LC}_3]$ in $\text{DMSO}-d_6$, and (c) composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ in $\text{DMSO}-d_6$. (d) ESI-MS spectrum of $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$.

teraction of $[\text{Zn}_3\text{LC}_3]^{6+}$ and $[\text{Mo}_6\text{O}_{19}]^{2-}$. Composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ was further characterized by the electrospray ionization mass spectrometry (ESI-MS) to determine the molecular composition. In Figure 1d, a set of peaks with different charge states (4+ and 6+) were observed (losing two or three $[\text{Mo}_6\text{O}_{19}]^{2-}$). The isotope pattern of each peak (Figures 1d and S26) agreed well with the corresponding simulated isotope pattern of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ with a molecular weight of 5514.81 Da, proving the composition of one $[\text{Zn}_3\text{LC}_3]^{6+}$ and three $[\text{Mo}_6\text{O}_{19}]^{2-}$. It is worth noting that there are no signals with odd charges in ESI-MS, indicating the PF_6^- was completely replaced by $[\text{Mo}_6\text{O}_{19}]^{2-}$.

Morphologies of the composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$

The powder of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ dispersed in acetonitrile was drop-cast onto a carbon-coated Cu micro-grid for transmission electron microscope (TEM) measurements. As shown in Figure 2, the lamellar structure was observed by TEM. The high-resolution transmission electron microscope (HRTEM) image showed a fine parallel stripe substructure (Figure 2b) due to the strong contrast^[40] between $[\text{Mo}_6\text{O}_{19}]^{2-}$ and $[\text{Zn}_3\text{LC}_3]^{6+}$, in which the layer spacing was 2.0 nm. The elemental mappings indicated the presence of N, Mo, and Zn elements in the lamellar structures, which provided additional evidence for the successful formation of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$.^[41] The SEM images (Figures 2h and 2i) further confirmed that the stacking mode was layer by layer. Moreover, the edge of each layered structure consisted of sharp serrations with 45° angle.

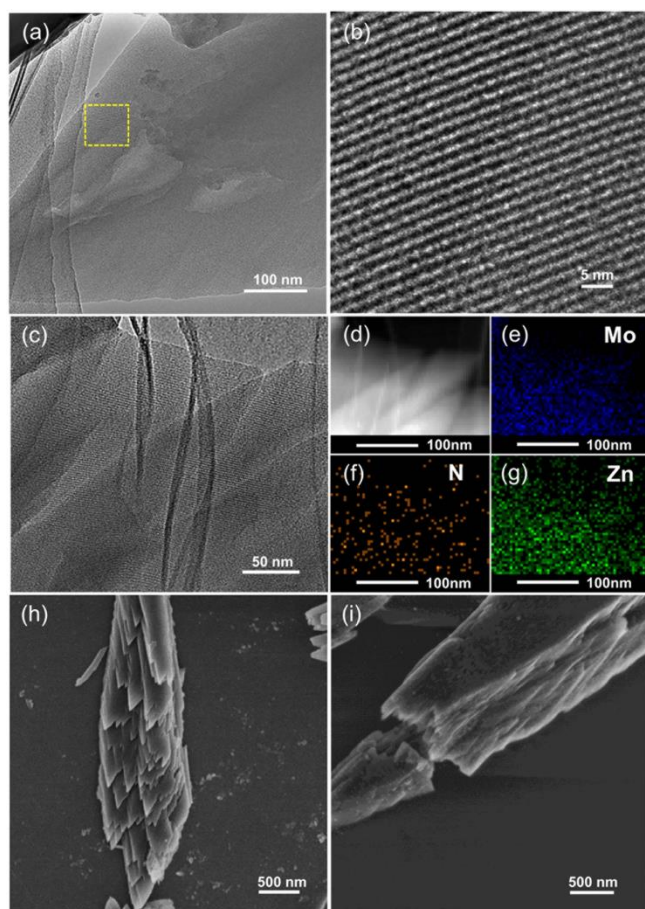


Figure 2 (a) and (c) TEM images of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ dispersed in acetonitrile. (b) Magnified TEM image of the area marked in Figure 2a. (d) HAADF STEM image and (e–g) the elemental mappings of selected area in Figure 2c. (h) and (i) SEM images of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ dispersed in acetonitrile.

X-ray structure

To further understand the formation mechanism of this composite, its single crystal was successfully obtained by slow vapor diffusion of acetonitrile into *N*-methyl pyrrolidone (NMP) solution containing composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ for four weeks. This crystal showed light yellow parallelograms with an angle of 45° (Figure S33), which agreed well with the angle of sharp serration edges observed by SEM images (Figure 2h). X-ray crystal structure of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ revealed that there were three $[\text{Mo}_6\text{O}_{19}]^{2-}$ anions around one triangle skeleton $[\text{Zn}_3\text{LC}_3]^{6+}$ (Figure 3). To accommodate $[\text{Mo}_6\text{O}_{19}]^{2-}$ anions, the cationic triangle skeleton showed slight distortion in the crystal structures. The two sides of the triangle skeleton were bent downwards at angles of 153.33° and 157.87°, respectively, and the third side was bent inward at angles of 173.28° (Figures S34a and 34b). Another triangle was symmetrical about the origin O, like two bowls arranged bottom to bottom (Figure S34d). As shown in the crystal structure, enough spaces were created and the anions were arranged not only in the bowl but also in the space between the two bowls. The $[\text{Mo}_6\text{O}_{19}]^{2-}$ anions were located not directly above or below the Zn(II), but at the three vertices of the triangle skeleton as expected. The reason could be that the vertical coordination between terpyridine and Zn(II) causes a certain steric hindrance to prevent large anions from approaching. The formation of two triangle skeletons and six $[\text{Mo}_6\text{O}_{19}]^{2-}$ anions unit mainly derived from electrostatic interactions and multiple C—H···O hydrogen-bonding interactions^[42] (Figure S34). The unit further interacted with its adjacent units based on electrostatic interactions, C—H···O, and π - π interactions to form porous structure.

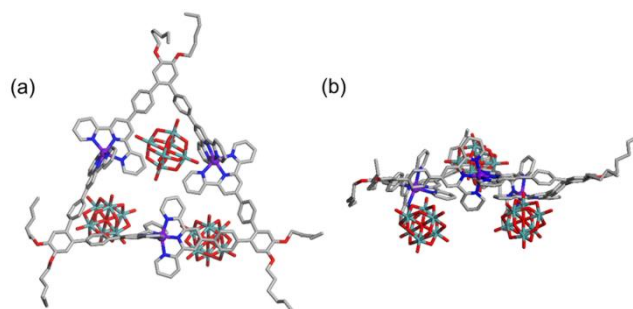


Figure 3 X-ray crystal structure of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$. (a) Top view, (b) side view. Color code for C: gray, N: blue, O: red, Zn: purple, Mo: green.

Within this framework, irregular micropore channels along the Y and X axis (Figures 4a and 4b) can be discerned, where the distances of the narrowest point were 1.9 Å and 2.4 Å, respectively, and the thickest part was 8.0 Å. According to other reported microporous materials, this framework has great potential to absorb volatile organic compounds (VOCs).^[43] As shown in Figure 4c, the sizes of benzene and toluene matches well with that of micropore channels. Therefore, benzene, toluene, and cyclohexane were selected for vapor sorption experiments, in which cyclohexane with large steric hindrance was used as a control. To adjust the sizes of pores, another three composites $[\text{Zn}_3\text{LA}_3][\text{Mo}_6\text{O}_{19}]_3$, $[\text{Zn}_3\text{LB}_3][\text{Mo}_6\text{O}_{19}]_3$, and $[\text{Zn}_3\text{LD}_3][\text{Mo}_6\text{O}_{19}]_3$ with different alkyl chains were synthesized (Scheme S1).

Vapor adsorption properties of four composites

The capture capacity of four composites for benzene, toluene, and cyclohexane was evaluated by vapor sorption isotherms. Composites $[\text{Zn}_3\text{LA}_3][\text{Mo}_6\text{O}_{19}]_3$ and $[\text{Zn}_3\text{LB}_3][\text{Mo}_6\text{O}_{19}]_3$ without long alkyl chains can absorb more VOCs than $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ and $[\text{Zn}_3\text{LD}_3][\text{Mo}_6\text{O}_{19}]_3$ at 298 K (Figure 5), indicating that the

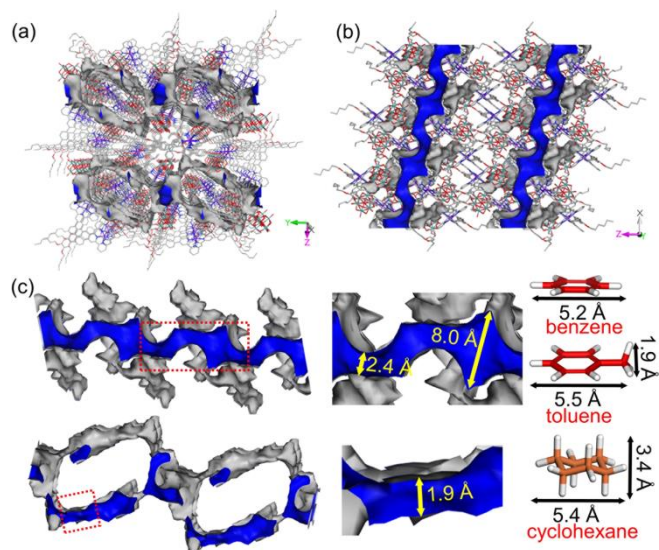


Figure 4 (a) Top view and (b) side view of packing diagram and channel of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$, with the solvent-accessible void space visualized by blue/gray (inner/outer) curved planes generated with a probe of 1.4 Å. (c) Schematic diagram of the simulated cross-section and VOCs.

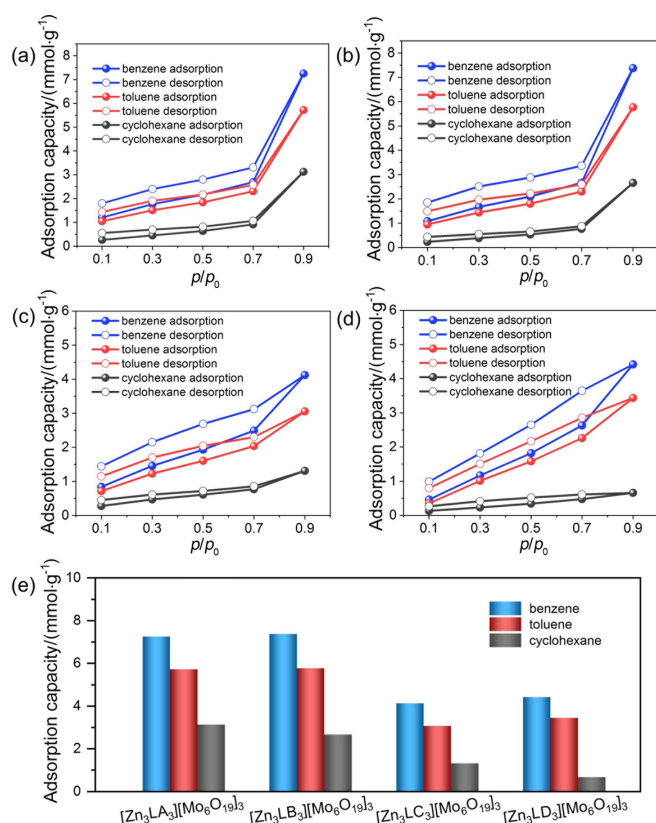


Figure 5 Vapor sorption isotherms of composites $[\text{Zn}_3\text{LA}_3][\text{Mo}_6\text{O}_{19}]_3$ (a), $[\text{Zn}_3\text{LB}_3][\text{Mo}_6\text{O}_{19}]_3$ (b), $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ (c), and $[\text{Zn}_3\text{LD}_3][\text{Mo}_6\text{O}_{19}]_3$ (d) toward benzene (red triangles), toluene (blue circles), and cyclohexane (black circles). Solid symbols: adsorption. Open symbols: desorption. (e) Adsorption capacity of benzene, toluene and cyclohexane (298 K, $p/p_0 = 0.9$).

presence of long alkyl chains may occupy space of pores and reduce the adsorption capacity for VOCs. Moreover, these four composites all showed different adsorption capacities increasing

in the order cyclohexane < toluene < benzene (298 K, $p/p_0 = 0.9$). Among them, the adsorption capacity of benzene was only slightly higher than that of toluene, which was much higher than that of cyclohexane. The reason may be that these VOCs exhibit distinct spatial configurational orientations;^[44] that is, benzene and toluene showed a planar π -cloud entity that can be easily absorbed, while cyclohexane exhibited a stable chair configuration with a width of 3.4 Å that hindered the adsorption (Figure 4c). It is worth mentioning that as the proportion of alkyl chains in the framework increases, the ratio of adsorption capacity between benzene and cyclohexane increases, indicating that this type of ionic framework has great potential to separate VOCs by reasonable design in the future.

Conclusions

In summary, a series of novel ionic complexes were constructed using metallo-triangles and polyoxometalates. Due to the strong electrostatic interactions, the products were rapidly formed (within 1 min) under room temperature and maintained high stability. The obtained composites $[\text{Zn}_3\text{LA}_3][\text{Mo}_6\text{O}_{19}]_3$, $[\text{Zn}_3\text{LB}_3][\text{Mo}_6\text{O}_{19}]_3$, $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$, and $[\text{Zn}_3\text{LD}_3][\text{Mo}_6\text{O}_{19}]_3$ with different alkyl chains were fully characterized by NMR and ESI-MS. Moreover, lamellar structures and good parallel stripe substructures of composite $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ were observed by TEM, HRTEM, and SEM. Further, the X-ray crystal structure of $[\text{Zn}_3\text{LC}_3][\text{Mo}_6\text{O}_{19}]_3$ showed 3D precise structure with multiple micropores, which enable its applications in vapor adsorption. The four composites showed differential adsorption selectivity toward benzene, toluene and cyclohexane. This work provides a novel approach for fabricating porous materials through electrostatic interactions, which display high potential for gas separation, energy storage, ion exchange, and heterogeneous catalysis.

Experimental

Materials

All reagents were purchased from Ark Pharm and J&K without further purification. Column chromatography was conducted using SiO_2 (VWR, 40–60 μm , 60 Å) and the separated products were visualized by UV light.

Nuclear magnetic resonance (NMR)

NMR spectra data were recorded on Bruker 600 MHz, 500 MHz, and 400 MHz nuclear magnetic resonance instruments using CDCl_3 , $\text{DMSO}-d_6$, and $\text{DMF}-d_7$ as the solvents.

Electrospray ionization-mass spectrometry (ESI-MS)

Both spectra were collected on a Waters Synapt G2 tandem mass spectrometer, using solutions of 0.5 mg sample in 1 mL of acetone/ CH_3OH (3 : 1, V/V) for composites.

Transmission electron microscope (TEM)

The powder of composites was dispersed in acetonitrile sonicated for 7 min, and the mixed solution was drop cast on an ultra-thin carbon-coated Cu grid (400 mesh) to a carbon-coated Cu micro grid for TEM measurements. The TEM images were taken with JEM-2100F and Hitachi-7700 transmission electron microscopes.

Scanning electron microscope (SEM)

The powder of composites was dispersed in acetonitrile and sonicated for 7 min. The mixed solution was drop cast on the silicon substrate for SEM measurements. The SEM images were taken with HITACHI SU8020 scanning electron microscope.

Thermogravimetric analysis (TGA)

TGA data were collected on a Q500 Thermal Analyzer in flowing N₂ with a heating rate of 10 °C·min⁻¹ in the temperature range from 30 °C to 900 °C.

Vapor sorption

Benzene, toluene, and cyclohexane vapors adsorption-desorption isotherms were collected on a BSD-VVS weight method analyzer at 25 °C. Samples were degassed under dynamic vacuum for 12 h at 80 °C prior to each measurement.

Supporting Information

The supporting information for this article is available on the WWW under <https://doi.org/10.1002/cjoc.202100798>.

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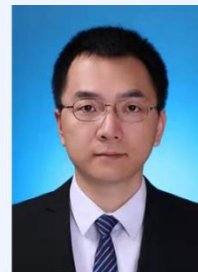
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Left to Right: (Top) Ming Wang, Lixin Wu, Bao Li, and Lu Li; (Bottom) Yaping Xu, Hao Yu, Xin Jiang, and Junjuan Shi.